



CS306 – Programming using Python

Assignment No.1 Solution

Semester: FALL 2025

Total Marks = 20

Deadline
05th of Nov 2025

Please carefully read the following instructions before attempting the assignment Solution.

NOTE

Don't copy-paste the same answer.

Make sure you can make some changes to your solution file before submitting copy paste solution will be marked zero.

If you found any mistake then correct yourself and inform me.

Before submitting an Assignment GDB checks your assignment requirement file.

For any query, feel free to Contact at
WhatsApp: +923074960034

Provide by **M.JUNAID QAZI**

Code:

```
def calculate_bill(name, ctype, units):
```

```
    ctype = ctype.strip().upper()
```

```
    rate_R = 10.5
```

```
    gst_R = 15.75
```

```
    rate = 0.0
```

```
    gst = 0.0
```

```
    discount = 0.0
```

```
    if ctype == 'R':
```

```
        rate = rate_R
```

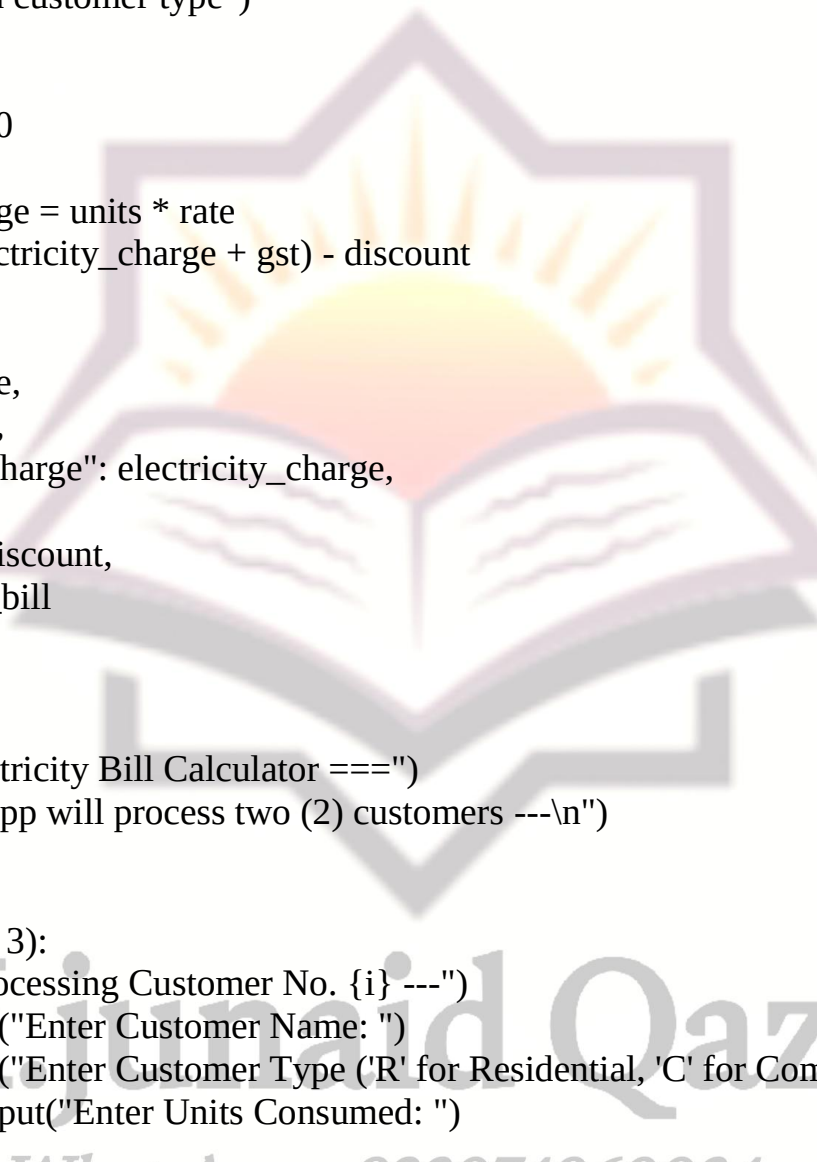
```
        gst = gst_R
```

```
        if units <= 200:
```

```
            discount = 15.0
```

```
    elif ctype == 'C':
```

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```

rate = rate_R * 2
gst = gst_R * 2
if units <= 300:
    discount = 30.0
else:
    print("Invalid customer type")
    rate = 0.0
    gst = 0.0
    discount = 0.0

electricity_charge = units * rate
total_bill = (electricity_charge + gst) - discount

return {
    "name": name,
    "units": units,
    "electricity_charge": electricity_charge,
    "gst": gst,
    "discount": discount,
    "total": total_bill
}

def main():
    print("=== Electricity Bill Calculator ===")
    print("--- This app will process two (2) customers ---\n")

    results = []
    for i in range(1, 3):
        print(f"--- Processing Customer No. {i} ---")
        name = input("Enter Customer Name: ")
        ctype = input("Enter Customer Type ('R' for Residential, 'C' for Commercial): ")
        units_str = input("Enter Units Consumed: ")
        try:
            units = float(units_str)
        except ValueError:
            units = 0.0
        print()
        bill = calculate_bill(name, ctype, units)

```

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```

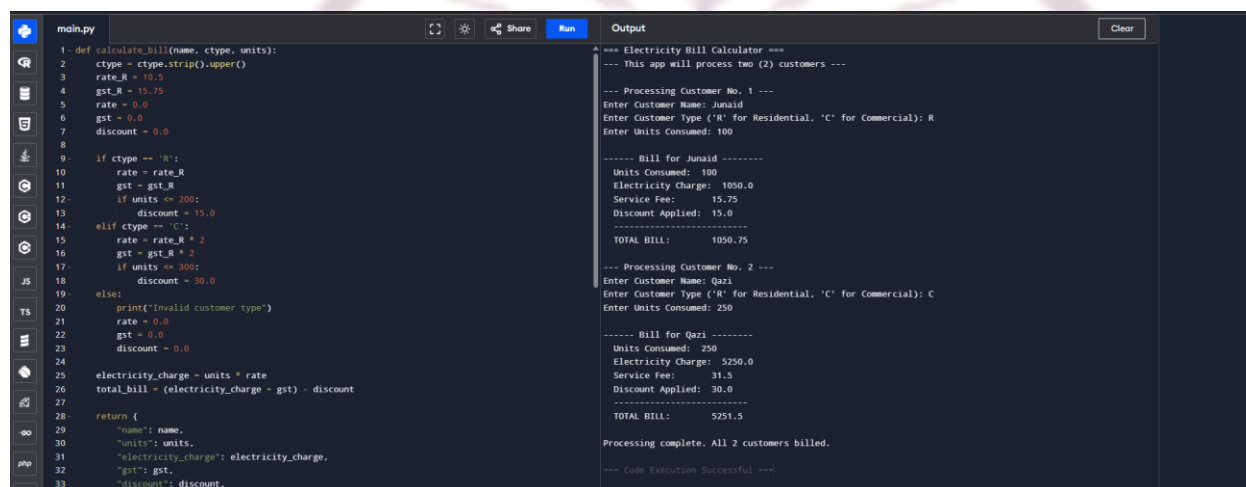
results.append(bill)
print(f"----- Bill for {bill['name']} -----")
units_display = int(bill['units']) if bill['units'].is_integer() else bill['units']
print(f" Units Consumed: {units_display}")
print(f" Electricity Charge: {bill['electricity_charge']}")
print(f" Service Fee: {bill['gst']}")
print(f" Discount Applied: {bill['discount']}")
print(f" -----")
print(f" TOTAL BILL: {bill['total']}\n")

print(f"Processing complete. All {len(results)} customers billed.")

if __name__ == "__main__":
    main()

```

OUTPUT:



```

main.py
1 def calculate_bill(name, ctype, units):
2     ctype = ctype.strip().upper()
3     rate_R = 10.5
4     gst_R = 15.75
5     rate = 0.0
6     gst = 0.0
7     discount = 0.0
8
9     if ctype == 'R':
10        rate = rate_R
11        gst = gst_R
12        if units <= 200:
13            discount = 15.0
14    elif ctype == 'C':
15        rate = rate_R * 2
16        gst = gst_R * 2
17        if units <= 300:
18            discount = 30.0
19    else:
20        print("Invalid customer type")
21        rate = 0.0
22        gst = 0.0
23        discount = 0.0
24
25    electricity_charge = units * rate
26    total_bill = (electricity_charge + gst) - discount
27
28    return {
29        "name": name,
30        "units": units,
31        "electricity_charge": electricity_charge,
32        "gst": gst,
33        "discount": discount,

```

```

==== Electricity Bill Calculator ====
--- This app will process two (2) customers ---

--- Processing Customer No. 1 ---
Enter Customer Name: Junaid
Enter Customer Type ('R' for Residential, 'C' for Commercial): R
Enter Units Consumed: 100

----- Bill for Junaid -----
Units Consumed: 100
Electricity Charge: 1050.0
Service Fee: 15.75
Discount Applied: 15.0
-----
TOTAL BILL: 1050.75

--- Processing Customer No. 2 ---
Enter Customer Name: Qazi
Enter Customer Type ('R' for Residential, 'C' for Commercial): C
Enter Units Consumed: 250

----- Bill for Qazi -----
Units Consumed: 250
Electricity Charge: 5250.0
Service Fee: 31.5
Discount Applied: 30.0
-----
TOTAL BILL: 5251.5

Processing complete. All 2 customers billed.

--- Code Execution Successful ---

```

Every Assignment/GDB is change due to unique Student ID so don't copy
That is truly perfect step by step idea solution get help easily.

Wish you the very best of Luck!4

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